UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Class 014 · Applications of LDEs: LCR circuits · compiled 17 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S014·8	Unit 2 deck 014 ("Applications of DE – LCR circuit"), slide 8 (rendered .pptx)
T014 00:04:29	Unit 2 AV-summary transcript "Applications of higher-order linear differential equations – RLC Circuits" (11.6 min), segment start
CW U2 P34	Unit 2 class-work problem 34
ESA Jan-24 s3 3b	Past paper: UE23MA141A Jan 2024 set 3, question 3.b

Contents

1. 1 · Modelling the series LCR circuit
2. 2 · Solving the circuit equation: CF, PI and damping
3. 3 · Past-paper circuits: AC sources and free discharge
4. 4 · First-order circuits: RL and RC
5. Past-paper questions

1 Modelling the series LCR circuit

Definitions

Series LCR (RLC) circuit. An inductor of inductance L (henries), a capacitor of capacitance C (farads) and a resistor of resistance R (ohms) connected in series with a source of electromotive force $E(t)$ (volts). S014·4 · T014 00:00:13

Current $I(t)$. The current through the whole circuit, in amperes; $I = \frac{dQ}{dt}$. S014·4 · S014·6 · T014 00:02:38

Charge $Q(t)$. The charge on the capacitor, in coulombs. S014·4 · S014·6

Voltage drop. The potential difference across one component: $L\frac{dI}{dt}$ (inductor), RI (resistor, Ohm's law), $\frac{Q}{C}$ (capacitor). S014·5 · T014 00:01:18

Kirchhoff's voltage law. The voltage supplied by the battery equals the sum of the voltage drops across the circuit components. S014·5 · T014 00:01:51

Initial conditions. The charge $Q(0)$ and the current $I(0) = Q'(0)$ at $t = 0$; they fix c_1 and c_2. S014·7 · T014 00:03:41

Statements in the class material

> *"A simple RLC circuit consists of an inductor of inductance 'L' (in henries), a Capacitor of capacitance 'C' (in farads) and a resistor of resistance 'R' (in ohms) and a source of electromotive force E(t) (in volts) are connected in series. The current 'I' through the circuit is measured in amperes and the charge 'Q' on the capacitor is measured in coulombs."* S014·4

> *"From Kirchoff's Voltage Law, The voltage supplied by the battery = The voltage drop across the circuit components (the sum of these voltage drops is equal to the electromotive force)."* S014·5

> *"$L\frac{d^2Q}{dt^2} + R\frac{dQ}{dt} + \frac{Q}{C} = E(t)$. This is the differential equation which describes LCR – Circuit."* S014·6

> *"It is a second order linear differential equation with constant coefficients. If the charge and the current are known at zero time, we can get an expression for Q(t) and I(t)."* S014·7

Key points

1. Three drops, one law: $L\frac{dI}{dt} + RI + \frac{Q}{C} = E(t)$; then substitute $I = \frac{dQ}{dt}$ to get one equation in Q. ESA ★★ S014·5 · S014·6 · T014 00:02:38
2. The LCR equation is a second-order non-homogeneous linear DE with constant coefficients: solve it with CF + PI. ESA ★★ S014·7 · T014 00:03:21 · T014 00:06:03
3. Two constants need two conditions: the charge and the current at $t = 0$. ESA ★★ S014·7 · T014 00:03:41
4. Substitute the numbers first and divide by L: $2Q'' + 8Q' + \frac{Q}{0.1} = 10 \Rightarrow Q'' + 4Q' + 5Q = 5$. ESA ★★ S014·8 · S014·9 · T014 00:05:46
5. Watch $\frac{Q}{C}$: $\frac{1}{0.1} = 10$, $\frac{1}{16\times10^{-4}} = 625$, $\frac{1}{0.002} = 500$, $\frac{1}{10^{-3}} = 1000$. ESA ★★ S014·9 · ESA Jan-24 s3 3b · ESA Jan-24 s4 3b
6. To get the current, differentiate the final $Q(t)$; or solve E1.5 for I directly (W1.1). ESA ★★ S014·12 · T014 00:10:31

Why?

Why does $I = \frac{dQ}{dt}$ turn E1.2 into a second-order equation?
$L\frac{dI}{dt}$ becomes $L\frac{d^2Q}{dt^2}$, RI becomes $R\frac{dQ}{dt}$, and $\frac{Q}{C}$ stays: one unknown, Q, with derivatives up to order 2.

S014·6 · T014 00:02:58

Exam traps

Exam trap. $\frac{Q}{0.1} = 10Q$, not $0.1Q$; and $\frac{1}{16\times10^{-4}} = 625$, not $\frac{1}{16}$. S014·9 · ESA Jan-24 s3 3b

Exam trap. Divide the whole equation by L, including $E(t)$: $Q'' + 4Q' + 5Q = 5$ (not 10). S014·9 · T014 00:05:46

2 Solving the circuit equation: CF, PI and damping

Definitions

Transient. The CF part of $Q(t)$; it decays to zero because every root has a negative real part. ADDED FOR CLARITY S014·10 · S014·16

Steady state. The PI part, which remains for large t: CE for a constant source, a sinusoid at the source frequency for an AC source. ADDED FOR CLARITY S014·10 · S014·15

Overdamped, critically damped, underdamped. $R^2 > \frac{4L}{C}$ (two real roots, no oscillation), $R^2 = \frac{4L}{C}$ (equal roots), $R^2 < \frac{4L}{C}$ (complex roots, decaying oscillation). ADDED FOR CLARITY S014·9 · S014·15

Key points

1. $Q(0)$ gives c_1 at once; for c_2 differentiate $Q(t)$ (product rule on $e^{-2t}(\dots)$) and use $I(0)$. ESA ★★ S014·11 · S014·12 · T014 00:09:00
2. For a battery (E constant) the PI is a constant: replace D by 0, giving $Q_{ss} = CE$, and $I \to 0$. ESA ★★ S014·10 · T014 00:07:52
3. R^2 against $\frac{4L}{C}$ decides the CF: real distinct (overdamped), equal (critical), complex (underdamped). Slide 014·8: $64 < 80$, underdamped; slide 014·13: $10^4 > 400$, overdamped. ESA ★ ADDED FOR CLARITY S014·9 · S014·15
4. The CF always decays (all roots have negative real part when $R > 0$), so it is the transient; the PI is what you measure after a while. ESA ★★ ADDED FOR CLARITY S014·16
5. With an AC source the steady state has the source frequency, not ω_d; its amplitude is largest near $\omega = \frac{1}{\sqrt{LC}}$ (resonance). ESA ★ ADDED FOR CLARITY S014·15

Why?

Why does the transient always die out?
The roots of $Lm^2 + Rm + \frac{1}{C} = 0$ have sum $-\frac{R}{L} < 0$ and product $\frac{1}{LC} > 0$: both real roots are negative, or the complex pair has real part $-\frac{R}{2L} < 0$. So every CF term carries a decaying exponential.
ADDED FOR CLARITY S014·9 · S014·15

Why is the battery steady state $Q = CE$?
When nothing changes, $I = 0$ and $\frac{dI}{dt} = 0$, so E1.2 leaves $\frac{Q}{C} = E$: the capacitor is fully charged and blocks the current.
ADDED FOR CLARITY S014·10 · S014·12

Exam traps

Exam trap. $I(0) = 0$ is a condition on $Q'(0)$: differentiate $Q(t)$ (including the $-2e^{-2t}$ term of the product rule) before substituting $t = 0$. S014·11 · S014·12 · T014 00:09:19

Exam trap. Slide 014·11 writes $e^{-2.0}$: it means $e^{-2\cdot 0} = 1$, not e^{-2}. ADDED FOR CLARITY S014·11

Exam trap. Slide 014·16 uses $i(0) = 0$ and $i'(0) = 0$ but states only the first; $i'(0) = \ddot{q}(0) = 0$ comes from the equation at $t = 0$. ADDED FOR CLARITY S014·16

3 Past-paper circuits: AC sources and free discharge

Definitions

Free (natural) discharge. $E(t) = 0$: a charged capacitor discharges through L and R; the charge is the CF alone. CW U2 P36

Key points

1. ESA "find the charge at any time t" with no initial conditions: give $Q = \mathrm{CF} + \mathrm{PI}$ with c_1, c_2 arbitrary. ESA ★★ ESA Jan-24 s3 3b · ESA Jan-24 s4 3b
2. Both ESA circuits are underdamped: $m = -20 \pm 15i$ and $m = -10 \pm 20i$ (same as the class 002 pack). ESA ★★ ESA Jan-24 s3 3b · ESA Jan-24 s4 3b
3. PI of a sinusoid: $D^2 \to -\omega^2$, a lone D remains, rationalise with the conjugate (class 004). ESA ★★ ESA Jan-24 s3 3b · ESA Jan-24 s4 3b
4. Free discharge ($E = 0$, CW P36): $Q(0)$ is the initial charge (here 0.002 C), not 0. ESA ★★ CW U2 P36

Why?

Why can the Type 2 rule not fail in an LCR circuit with $R > 0$?
Failure needs $\pm i\omega$ to be AE roots, i.e. purely imaginary roots. With $R > 0$ the roots have negative real part, so $f(-\omega^2) \neq 0$.

added for clarity ESA Jan-24 s3 3b

Exam traps

Exam trap. With no initial conditions do not set $c_1 = c_2 = 0$: the answer keeps the transient with arbitrary constants. added for clarity ESA Jan-24 s3 3b · ESA Jan-24 s4 3b

Exam trap. CW P36's 0.0008 is $\frac{0.002}{\sqrt{7}} = 0.000756$ rounded; 1323 is $500\sqrt{7} = 1322.9$.

added for clarity CW U2 P36

4 First-order circuits: RL and RC

Definitions

Time constant. $\frac{L}{R}$ (RL) or RC (RC): the time in which the transient falls to $e^{-1} \approx 37\%$.
ADDED FOR CLARITY CW U2 P34 · CW U2 P35

Key points

1. RL: $L\frac{di}{dt} + Ri = E(t)$ is first order in i; RC: differentiate $Ri + \frac{Q}{C} = E$ to get $R\frac{di}{dt} + \frac{i}{C} = \frac{dE}{dt}$. ESA ★★ CW U2 P34 · CW U2 P35
2. A single root $m = -\frac{R}{L}$ or $-\frac{1}{RC}$ gives one constant, fixed by $i(0) = 0$. ESA ★★ CW U2 P34 · CW U2 P35

Exam traps

Exam trap. CW P34's printed answer has $e^{-Ri/L}$ and $\sin\varphi\varphi$: read $e^{-Rt/L}\sin\varphi$. CW P35's has $e^{-1/RC}$: read $e^{-t/(RC)}$. ADDED FOR CLARITY CW U2 P34 · CW U2 P35

★ Past-paper questions

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024 (19 distinct Unit 2 questions, 24 occurrences). LCR circuits: 2 distinct questions, 2 occurrences, both 6 marks and both in the Jan 2024 sets: set 3 Q3.b ($R = 40\,\Omega$, $L = 1$ H, $C = 16 \times 10^{-4}$ F, $E = 100\cos 5t$) and set 4 Q3.b ($R = 20\,\Omega$, $L = 1$ H, $C = 0.002$ F, $E = 12\sin 10t$), each "find the charge at any time t" with no initial conditions (★★). CW P34–P36 add the RL and RC circuits and a free LCR discharge (★★). Never asked: the derivation of the model as a question on its own, the current equation, the damping classification and the resonance/impedance view (★).

3 · Past-paper circuits: AC sources and free discharge

ESA Jan 2024 · 6 marks · UE23MA141A set 3, Q3.b · solved in W3.1

A series circuit has $R = 40\,\Omega$, $L = 1$ H, $C = 16 \times 10^{-4}$ F and $E = 100\cos 5t$. Find the charge at any time t.

ESA Jan 2024 · 6 marks · UE23MA141A set 4, Q3.b · solved in W3.3

A series circuit has $R = 20\,\Omega$, $L = 1$ H, $C = 0.002$ F and $E = 12\sin 10t$. Find the charge at any time t.